

Online Ensemble Learning for Load Forecasting

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Abstract—Traditionally, load forecasting models are trained offline and generate predictions online. However, the pure batch learning approach fails to incorporate new load information available in real-time. Conversely, online learning allows for efficient adaptation of newly incoming information. This letter advocates a novel online ensemble learning approach for load forecasting by combining batch and online learning. While the individual batch models provide an appropriate forecast basis, the online ensemble combines their predictions and ensures adaptivity for online application. In that respect, we propose a modified Passive Aggressive Regression (PAR) model to implement the online ensemble forecasting. Case studies on a real-world load dataset show that the proposed method can improve the forecasting accuracy significantly compared to a pure batch learning approach.

Index Terms—Load forecasting, online learning, ensemble learning, passive aggressive regression

I. INTRODUCTION

MOST of the load forecasting literature focuses on offline training on batch data and subsequent online forecasting. Popular models are Support Vector Regression (SVR), Random Forests (RF) or Gradient Boosting Regression Tree (GBRT) among others [1], [2], [3]. However, these batch models cannot be trained incrementally with new samples. Other works have considered a sliding window approach, predominantly with Artificial Neural Networks or incremental SVR [4], [5]. Hereby, a sample window is dynamically increased and decreased to allow an incremental learning approach. This strategy is certainly promising, but entails elevated computational burden due to the sample window. Nonetheless, very little work has been done on online learning. Although the widely used ANN would be capable of online learning, most works investigate its use in a batch context [6], [7]. Exceptions present Online Sequential Extreme Learning Machine (OSELM) in [8] or Gaussian Regression in [9]. Pure online machine learning features linear time complexity and constant space complexity. Therefore, it allows to incorporate new information very efficiently while having low computational burden. This is particularly interesting in two settings. One is the prediction of massive loads, where updating the forecasting models on batch data is time-consuming and storage space of the infrastructure is limited. The other setting is very short-term load forecasting, where model updates need to be carried out at a high frequency.

On one hand, batch load forecasting models are often elaborate and exhibit high accuracy on batch data. Furthermore,

a data batch is oftentimes available to the practitioner. On the other hand, online learning provides efficient adaptivity paramount for online application. In order to harness benefits of both paradigms, we propose a combination of batch and online learning by means of ensemble learning. While the individual batch models provide a solid forecast basis after batch training, the online ensemble combines their predictions in an online fashion to ensure integration of new observations. Actually, the concept of online ensemble learning has been introduced by [10]. A simple but very effective ensemble learning approach is weighted averaging. Hence, this letter proposes an online load forecasting method based on the weighted averaging ensemble model. In this context, we propose a modified online Passive Aggressive Regression (PAR) algorithm to implement the online ensemble learning strategy.

This letter addresses the following question: *Can online learning be efficiently integrated into existing structures and lead to an improvement in online forecasting accuracy?* In that respect, we make three contributions:

- 1) Enrich the research on online load forecasting since this topic has been rarely touched in the load forecasting area;
- 2) Propose a novel ensemble forecasting method based on a modified PAR algorithm that enables accurate batch and online learning;
- 3) Improve the forecasting accuracy significantly compared with traditional batch approaches which is verified by real-world case studies.

The remainder of this letter is organized as follows: Section II defines the online ensemble learning problem and proposes the online optimization algorithm. Section III conducts case studies and makes comparisons to verify the effectiveness of our proposed method. Finally, Section IV draws conclusions and gives an outlook on future work.

II. PROPOSED METHODOLOGY

A. Problem Statement

Model averaging is an ensemble learning technique where several individual models $f_1(\mathbf{x}), \dots, f_M(\mathbf{x})$ are combined into a new global model $G(\mathbf{x})$ by taking a weighted average

$$G(\mathbf{x}) = \sum_{m=1}^M w_m f_m(\mathbf{x}) \quad (1)$$

where w_m denotes the weight of the m -th individual model f_m , $\mathbf{x}$ the input features and M the number of individual models to be combined.

We can interpret this averaging process as a form of regression procedure. From this perspective, we can employ an arbitrary regression model as a combiner. Besides, the averaging can naturally be applied in an online setting. The

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individual models f_m are thereby not required to be online learners themselves. The proposed method can operate in batch and online mode. The batch mode is based on the optimization problem from [11] formulated as

$$\begin{aligned} \min_{\mathbf{w}} \quad & L(y - G(\mathbf{x})) \\ \text{s.t.} \quad & \sum_{m=1}^M w_m = 1, \quad w_m \geq 0, \quad m = 1, \dots, M \end{aligned} \quad (2)$$

where, the objective function L is the Mean Absolute Percentage Error (MAPE) of its argument and y is the actual response. Subsequently, the online mode starts with the weights from the batch mode and proceeds finding optimal weights according to the modified PAR variant introduced in the next subsection.

B. Online Learning Algorithm

Online Convex Optimization (OCO) is a unifying framework for the analysis and design of online algorithms. In this letter, we focus on linear predictors as in (1). We denote the predictions of the individual models at time t with $\mathbf{f}_t \in \mathbb{R}^M$. In this case, the online protocol [12] is described in **Algorithm 1**. The key of OCO is to define the convex loss function ℓ and weight update rule U .

Algorithm 1: Online Protocol

input: Initial model weights $\mathbf{w}_1 \in \mathbb{R}^M$,
convex loss function ℓ , weight update rule U
for $t = 1, 2, \dots$
 Calculate individual predictions $\mathbf{f}_t \in \mathbb{R}^M$
 Predict $\hat{y}_t = \mathbf{w}_t \cdot \mathbf{f}_t$
 Reveal true value $y_t \in \mathbb{R}$
 Calculate loss $\ell(y_t, \hat{y}_t)$
 Update model $\mathbf{w}_{t+1} = U(\mathbf{w}_t; \ell(y_t, \hat{y}_t))$
end

One of the most fundamental algorithms in OCO is the Online Gradient Descent (OGD) [12]. In fact, OGD corresponds to the well-known Stochastic Gradient Descent (SGD) in an online setting. Thereby, the observations are processed in a sequential manner and are not reused. The OGD weight update rule can be written as

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta_t \nabla_{\mathbf{w}_t} \ell(y_t, \mathbf{w}_t \cdot \mathbf{f}_t) \quad (3)$$

where η_t is the learning rate and $\nabla_{\mathbf{w}_t} \ell$ is the gradient of the loss function of interest.

In connection with gradient descent, [13] provided a more general view of deriving online updates. The objective is to seek new weights $\mathbf{w}$ of the form

$$\mathbf{w}_{t+1} = \arg \min_{\mathbf{w}} \left[d(\mathbf{w}, \mathbf{w}_t) + \eta_t \ell(y_t, \mathbf{w} \cdot \mathbf{f}_t) \right] \quad (4)$$

The update rule balances two terms: The first is the distance function d , that prevents information loss to the next round by keeping the new weights close to the old weights. The second term is the loss function ℓ , which forces the new weights to have a small error on the new sample thereby incorporating

new information. The OGD update from (3) can be viewed as a special case where d is the squared l_2 -norm. According to the Karush-Kuhn-Tucker (KKT) conditions, we can obtain the optimal $\mathbf{w}_{t+1}$ by solving:

$$\begin{aligned} \nabla_{\mathbf{w}} \left[\frac{1}{2} \|\mathbf{w} - \mathbf{w}_t\|^2 + \eta_t \ell(y_t, \mathbf{w} \cdot \mathbf{f}_t) \right] \\ = \mathbf{w} - \mathbf{w}_t + \eta_t \nabla_{\mathbf{w}} \ell(y_t, \mathbf{w} \cdot \mathbf{f}_t) = 0 \end{aligned} \quad (5)$$

Thus, we have the same update rule as in (3) when considering explicit updates:

$$\mathbf{w} = \mathbf{w}_t - \eta_t \nabla_{\mathbf{w}} \ell(y_t, \mathbf{w} \cdot \mathbf{f}_t) \Big|_{\mathbf{w}=\mathbf{w}_t} \quad (6)$$

Inspired by this insight, [14] proposed the Passive Aggressive Regression. It is a margin-based linear predictor, which uses the squared l_2 -norm as distance function and the ε -insensitive loss reading:

$$\begin{aligned} \mathbf{w}_{t+1} &= \arg \min_{\mathbf{w}} \left[\frac{1}{2} \|\mathbf{w} - \mathbf{w}_t\|_2^2 + C\xi \right] \\ \text{s.t.} \quad & \ell_\varepsilon(y_t, \mathbf{w} \cdot \mathbf{f}_t) \leq \xi \\ \ell_\varepsilon(y_t, \mathbf{w} \cdot \mathbf{f}) &= \begin{cases} 0 & \text{if } |y - \mathbf{w} \cdot \mathbf{f}| \leq \varepsilon \\ |y - \mathbf{w} \cdot \mathbf{f}| & \text{otherwise} \end{cases} \end{aligned} \quad (7)$$

where ε is a constant to control the learning rate. In each update round, the algorithm either stays “passive” if the prediction is within the ε -margin, or corrects its weights “aggressively” in order to adapt to the new sample. Moreover, the parameter C allows to control the level of aggressiveness through the slack variable $\xi \geq 0$.

For high dimensional problems, the l_1 -norm is advocated as a preferable distance metric [15]. Thus, we propose a variant of the PAR model that uses the l_1 -norm as distance function. Besides, an l_1 -regularization term is added for the purpose of variance reduction and sparsity promotion. Finally, instead of setting the aggressiveness explicitly, we allow less aggressive behaviour by moving the ε -insensitive loss into the objective function. Thus, our proposed modified PAR model applies the following update rule:

$$\mathbf{w}_{t+1} = \arg \min_{\mathbf{w}} \left[\|\mathbf{w} - \mathbf{w}_t\|_1 + \ell_\varepsilon(y_t, \mathbf{w} \cdot \mathbf{f}_t) + \lambda \|\mathbf{w}\|_1 \right] \quad (8)$$

III. CASE STUDIES

A. Experiment Setup

In this letter, we focus on the practical scenario where we have access to a batch of historical data and want to start online learning on incoming samples. In our case study, we used the aggregated residential load data from the Smart Metering Electricity Customer Behaviour Trials [16] for this purpose. It contains electricity consumption recorded in half-hour intervals from July 20, 2009 until December 26, 2010.

The case study procedure is shown in Fig. 1 and can be divided into three stages. In the first stage, the data is partitioned into three sets $\mathbf{D}_{ind}$, $\mathbf{D}_{ens}$ and $\mathbf{D}_{test}$ of size 50%, 26% and 24%. While $\mathbf{D}_{ind}$ and $\mathbf{D}_{ens}$ serve as historical datasets, $\mathbf{D}_{test}$ is used to simulate an online learning scenario. In the second stage, the individual models are trained on $\mathbf{D}_{ind}$ and used to generate predictions for $\mathbf{D}_{ens}$. The ensemble model is then trained in batch mode on these individual predictions

and D_{ens} . In the third stage, online learning is performed on D_{test} in a streaming fashion. Hence, the ensemble is switched into online mode. The individual batch models first generate a prediction on an incoming sample. On this basis, the online ensemble model then generates a prediction and updates itself according to the online protocol. We use a total of 12 frequently used forecasting models including RF, GBRT and SVR as the individual models to be combined.

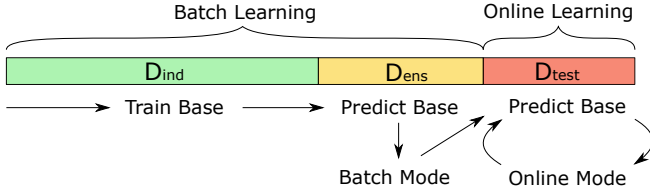


Fig. 1: Procedures and data partition of the case study.

The proposed method is compared to the Stochastic Gradient Descent Regressor (SGDR) with the least squares loss, the Passive Aggressive Regressor from the sci-kit learn package [17], the Follow-the-Regularized-Leader-Proximal (FTRL) [18], the Online Sequential Extreme Learning Machine (OSELM) [19] and Online Bagging [10] with SGDR models. All methods can operate in batch and online mode. With exception of OSELM and Online Bagging, the batch mode for the remaining methods is implemented as a stochastic subsampling procedure.

The hyperparameters are determined by a regular train-validation procedure on D_{ens} . The PAR and our proposed method were found to be robust to the choice of hyperparameters, i.e., the value of hyperparameters have little impact on the forecasting performance. We have therefore taken the standard values given in Table I. Regarding the OSELM, the sigmoid activation function was chosen. In order to simulate an online setting for SGDR, the learning rate η_t is fixed for online learning.

TABLE I: Hyperparameters for batch and online learning

Method	Batch Parameter	Online Parameter
SGDR	$\eta_t = 0.01 \cdot t^{-1/4}$	$\eta_t = 0.01$
FTRL	$\alpha = 0.05, \beta = 1.0, \lambda = 0.1$	
OSELM	layers = 1, neurons = 50	
Online Bagging	models = 100	
PAR	$\varepsilon = 0.1, C = 1.0$	$\varepsilon = 0.1, C = 1.0$
Proposed	-	$\varepsilon = 0.1, \lambda = 0.1$

B. Experimental Results

The predictive performances are evaluated with the four metrics MAPE, Standard Deviation (SD) of the Absolute Percentage Error (APE), Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). In Table II, the ensemble accuracies on D_{Test} are shown after batch learning in the second stage. The best individual batch model of each individual model type is added as a benchmark. We observe that our proposed method is the only ensemble having higher accuracy than the best individual models. In terms of stability, FTRL

and PAR and our method yield the lowest SD. Consequently, the proposed method displays the best performance in the case of pure batch learning evaluation.

TABLE II: Errors on test set after batch learning

Method	MAPE	SD	MAE	RMSE
SGDR	3.23%	0.030	125.74	180.18
FTRL	3.13%	0.028	113.70	161.08
OSELM	3.21%	0.030	125.18	185.14
Online Bagging	3.57%	0.030	122.37	165.12
PAR	2.94%	0.028	110.09	161.14
Proposed	2.89%	0.028	107.55	154.88
Best SVR	3.18%	0.032	117.54	171.72
Best RF	2.89%	0.029	108.25	156.84
Best GBRT	3.53%	0.032	127.81	175.78

The performance after online learning in the third stage is indicated in Table III. The ensemble methods are compared to the best individual batch models, the stand-alone optimization problem of (2) referred to as Batch OPT and a periodically retrained optimization problem denoted as Window OPT. The retraining is carried out in a sliding window manner with a window size of D_{ens} and a step size of one week. We identify three main tendencies in Table III. Firstly, all ensembles improve their forecasting performance through online learning. Secondly, nearly all ensembles outperform the benchmarks after online learning. An exception presents OSELM concerning SD and RMSE. In terms of MAPE, the performance discrepancy between the best ensemble and Window OPT is more than 1%. Thirdly, the proposed method has the highest accuracy and stability among all examined ensembles. This reveals that in comparison to pure batch models, a substantial performance improvement can be achieved by ensembles incorporating online learning. Furthermore, OSELM demonstrated only minor improvements and is therefore not particularly suited as online combiner.

TABLE III: Errors on test set after online learning

Method	MAPE	SD	MAE	RMSE
SGDR	2.43%	0.025	86.05	122.71
FTRL	2.23%	0.021	81.09	113.87
OSELM	2.80%	0.029	106.03	155.03
Online Bagging	2.07%	0.021	74.33	106.23
PAR	1.67%	0.015	61.83	86.68
Proposed	1.62%	0.014	59.59	83.21
Best SVR	3.18%	0.032	117.54	171.72
Best RF	2.89%	0.029	108.25	156.84
Best GBRT	3.53%	0.032	127.81	175.78
Batch OPT	2.89%	0.028	107.55	154.88
Window OPT	2.85%	0.028	106.13	149.81

From a practical perspective, it is also important to identify from which point on the online ensemble consistently outperforms traditional batch models. To this end, a direct comparison of running metrics for each incoming sample of D_{test} is carried out with Window OPT as a benchmark. The corresponding hour of outperformance is given in Table IV. Starting from the break-even hour, the online model consistently displays a lower error metric than the benchmark until the end of D_{test} . The results show that the proposed

method has the earliest break-even within 2 hours for all metrics. Almost all other ensembles achieve a break-even for all metrics approximately within one or two days. OSELM exhibits an extremely late MAE break-even and no RMSE break-even at all. Disregarding OSELM, this demonstrates that an ensemble employing online learning is able to pay off at a relatively early point in time. An illustrative example of the running MAPE over the first day of D_{test} is shown in Fig. 2.

TABLE IV: The hour of break-even for all ensembles

Method	Break-even [hour]			
	MAPE	SD	MAE	RMSE
SGDR	39.5	86.5	41.0	64.0
FTRLP	66.5	87.0	64.0	60.5
PAR	17.5	9.0	19.5	17.5
OSELM	112.0	2.0	2833.5	no
Online Bagging	22.5	4.5	23.0	35.5
Proposed	1.5	2.0	1.5	1.5

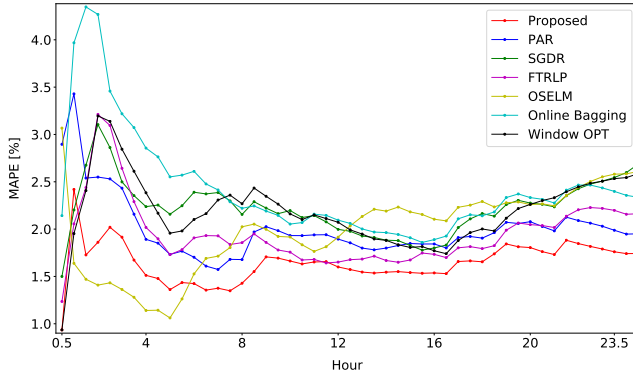


Fig. 2: MAPE over the course of the first day of online learning.

IV. CONCLUSIONS AND FUTURE WORKS

We demonstrated that online learning can be a powerful tool in short-term load forecasting. The main advantage is higher accuracy than traditional batch models already after several hours of online forecasting. Furthermore, we showed that our proposed modified PAR model is very suitable in this context, especially as an online ensemble method. In the future, we will extend the online ensemble learning from deterministic forecasting to probabilistic forecasting. The application of our proposed method to wind or solar forecasting will also be investigated.

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